

Figure 1: Circuit diagram of a DOT MATRIX

Task 1: Understand the basic configuration of 8255 and the DOT matrix.

LEDs are a particular type of diode that converts electrical energy into light. In fact, LED stands for “Light Emitting Diode”. The following figure is taken from TechTerms.com and shows the basic forward bias electricity flow from Anode (Positive +) to Cathode (Negative -). If both side of diode have same voltage value (1/0) then no conduction, means no current. However, if potential difference is equal to or greater than threshold (0.7 for germanium) then there will be conduction.

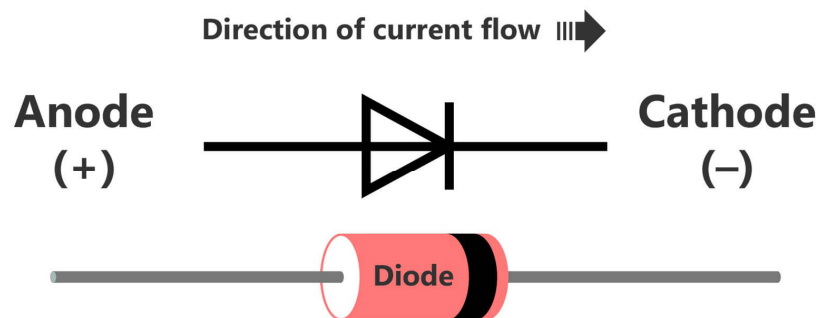


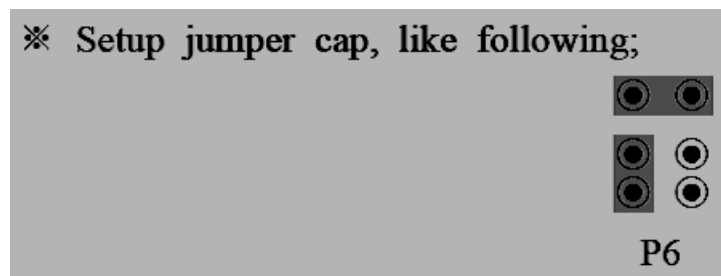
Figure 2: Direction of Current Flow

For LEDs in MATRIX: Glow 1st ROW LEDs

You will need to select the column A0/B0 (which means A0/B0 is pulled low), and deselect other columns by blocking their ground paths (by pulling A1/B1 through A7/B7 pins to logic high). Now, the first column is active, and you will need to turn on the LEDs in the rows C0 through C7 of this column, which can be done by applying forward bias voltages (HIGH) to all rows.

Jumper Setup:

You need to set the jumper as shown below before running any program with MDA kit.



Task 2: Understand the Setting of the Jumper.

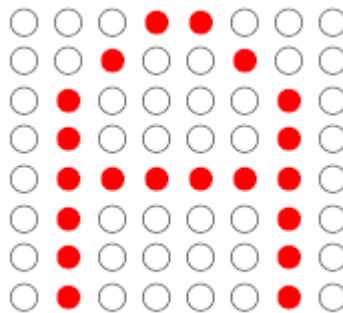
Go to top right corner of DOT MATRIX figure.

Experimental tools:

MDA-Win8086, Computer, Microprocessor emulator Software with Integrated Assembler.

The following rules are needed to perform the lab work.

Task 3: Run the HEX CODE of the following program to display the letter 'A' on the LED matrix.



```
; PPIC_C EQU 1EH ; control register
; PPIC EQU 1CH ; c port
; PPIB EQU 1AH ; b port
; PPIA EQU 18H ; a port
;
```

Address:	Op code	instructions
		ORG 1000H
1000:	B0 80	MOV AL, 10000000B
1002:	E6 1E	OUT PPIC_C, AL ; program PPI
		;
1004:	B0 FF	MOV AL, 11111111B ; OFF LEDs connected to port A
1006:	E6 18	OUT PPIA, AL
		;
1008:	BE 2C 10	L1: MOV SI, OFFSET FONT
		;
100B:	B4 01	MOV AH, 00000001B
		;
100D:	2E 8A 04	L2: MOV AL, BYTE PTR CS:[SI]
1010:	E6 1A	OUT PPIB, AL
		;
1012:	8A C4	MOV AL, AH
1014:	E6 1C	OUT PPIC, AL
1016:	E8 09 00	CALL TIMER

```

1019: 46          INC  SI
101A: F8          CLC
101B: D0 C4       ROL  AH, 1
101D: 73 EF(EF)  JNC  L2
101F: EB E7       JMP  L1
                        ;
1021: CC          INT  3
                        ;
1022: B9 2C 01    TIMER:  MOV  CX, 300
1025: 90          TIMER1: NOP
1026: 90          NOP
1027: 90          NOP
1028: 90          NOP
1029: E2 FA       LOOP TIMER1
102B: C3          RET
                        ;
102C: FF  FONT: DB  11111111B
102D: C0          DB  11000000B
102E: B7          DB  10110111B
102F: 77          DB  01110111B
1030: 77          DB  01110111B
1031: B7          DB  10110111B
1032: C0          DB  11000000B
1033: FF          DB  11111111B

```

Task 4: Write a program of display another letter